ARGUS: hardened prototype and technical review package

Security fixes, scientific integrity, fresh synthetic evaluation, publication scope and IP support.

Read this first

The code controls and reproducibility package have been strengthened and tested. The fresh stress study still contains large errors. This package makes the scope inspectable; it does not replace physical validation or establish novelty, filing status, acceptance or a course grade.

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Source documents and exact commands remain in docs/. Current JSON and JUnit evidence remain in output/release/ and research_results/. The separate manuscript PDF is compiled from paper/main.tex.

ARGUS security, scientific integrity and release report

Date: 7 October 2026. **Scope:** current local working tree, including edits that existed before this audit. Existing source history and September evidence were preserved. No external publication, patent filing, deployment or grade assessment is represented as completed.

The project is now a more defensible local research prototype with tested security and evidence controls, an updated manuscript and a practitioner-facing technical disclosure. It is not a field-validated NDE product or a demonstrated patentable invention. Some important scientific failures remain visible in the fresh study.

Executed checks

Check	Actual result	Evidence
Backend full suite in patched virtual environment	294 passed, 1 skipped; no failures	output/release/backend-tests.xml
Historical binary-model test	Skipped because legacy pickle deserialization is refused; safe-format prediction round trip passes	test_surrogate.py; test_safe_checkpoints.py
Frontend	26 tests passed; typecheck, lint and production build passed	Local commands and output/release/frontend-checks.json
Readiness	All imports, SQLite and one closed-loop measurement pass with PyTorch 2.6.0+cpu	scripts/doctor.py
Deterministic demo	Eight actions; approximately 10.0 mm MAP error; entropy 0.473; STOP_BUDGET	easy, seed17, eight observations
Fresh paired synthetic study	48/48 runs complete, source unchanged and rechecked against current backend	release_validation_2026_10_07.json
Release source/evidence bundle	Allowlisted members, hashes and declared evidence checked by builder/verifier	scripts/build_release_bundle.py

The one upstream Starlette/httpx test-client deprecation warning is retained; it is not a test failure. The main application source, regression tests and CLI workflows were verified locally; no full interactive browser or physical-device campaign was performed. Production build passes with Vinext's informational route-classification notice. A test or build is not proof that no vulnerability exists outside its declared cases.

Remediated security and control issues

Severity below is this audit's local threat assessment, not an invented CVSS score.

Finding	Risk	Implemented correction	Regression evidence
Anonymous LAN controls by default	High: session, jobs and physical controls exposed	Loopback default; explicit LAN mode, strong key, exact origins/hosts, HTTP/WS authentication	test_network_security.py

Finding	Risk	Implemented correction	Regression evidence
Hostile browser/socket origins and DNS-rebinding access	High: unauthorized local/LAN operation	Host, peer and exact-origin checks before routing/parsing	test_network_security.py
Hardware acquisition ran before latch/constraints	High: unsafe action could execute then be rejected	Validate stop, source geometry/no-go and drive before device.acquire; per-device serialization	Acquisition-spy security tests
HTTP/WS and job resource exhaustion	High: memory/CPU service denial	Byte/deadline/rate/concurrency/frame limits; bounded config and metadata; no automatic missing benchmark computation	Network/input/artifact tests
Unsafe pickle checkpoint readers; vulnerable PyTorch	High when an untrusted model is loaded	Remove torch.load readers; bounded ZIP/NPY/JSON format; patched project-local Torch2.6	Safe checkpoint tests; publisher advisory
Next.js dependency behind publisher security release	Potential affected-framework exposure	Next.js16.3.8 and updated lockfile; scripts/audit disabled during patch fetch	Frontend checks and build
Malicious ZIP/NPY/JSON/zlib artifacts	High: allocations/ambiguous state before validation	Preflight paths, headers, sizes, dtype, duplicates, expansion and stored decompression; strict JSON	test_artifact_security.py
Imported state did not bind enough content	Medium: inconsistent posterior/model/verification state	Validate complete transition chain and online buffers; schema2 planner/diagnostic hashes	Artifact and evidence tests
Commit-only evidence misidentified edited code	Medium: wrong reproducibility identity	Fingerprint actual tracked/untracked source bytes; retain independent head/count checkpoint option	Dirty-source and tampering tests
Edge key forwarded across redirects	High: operator credential disclosure	Redirect refusal and server-address validation	test_launcher_edge_security.py
Edge captured only its first recommendation	Medium: incomplete adaptive acquisition	Track the pre-capture count; register once per connection	Two-consecutive-capture regression
UI suppressed emergency stop while busy	High control defect	Independent stop dispatch, enabled stop button and preservation of acknowledged latch	Hook/panel review; backend stop regressions
Launcher could leave Node children and exposed key to frontend	Medium: orphan listener/secret exposure	Windows owned-process-tree termination; separate frontend environment	Launcher regression tests

Remediated scientific/evidence issues

Finding	Why it threatened interpretation	Correction
Sensor reliability evaluated after fusion	A faulty channel could influence the grid or calibration first	Gate reliability before structural/nuisance/discrepancy/screening updates
Short/2D/invalid signals silently changed observation	Missing or mixed evidence retained the original experiment label	Exact mono/window checks before engine mutation; reject missing data
Resampling hid original Nyquist failure	Interpolation cannot recover aliased measurement content	Require original-rate support for declared excitation band
Sensor name could suppress measured-reference OOD	Client labels changed whether physical evidence was checked	Server-owned physical-acquisition provenance
Raw physical unit labels used normalized-model thresholds	Unit names are not a transfer calibration	Explicit affine conversion and recorded reference; declaration remains unvalidated
Rejected measurements became repeat references on restart	Persistence could reintroduce poisoned consistency evidence	Restore only accepted, reliable, structurally supported or calibrated references
Probability overflow, negative/zero support, rounded persistence	Invalid support or tiny mass could silently change belief	Stable normalization/log update, support validation and full-precision state
Calibration used missing/unsupported diagnostics	Uncertainty could shrink without the required observation	Require valid measured diagnostics and support; no partial invalid update
Amplitude/duration/reciprocal changes counted as verification	No new structural path constraint	Meaningful geometry or excitation diversity required
Fixed severity prior influenced spatial uncertainty	Unlearned attributes changed a decision score	Location uncertainty uses learned spatial quantities; attributes labelled unchanged priors
Particle helper ignored speed/timing covariance	Inconsistent prediction sampling	Correlated sampling helper; explicitly not deployed hierarchical inference
Gain variance missing from predictive log gain	Mean depended on an uncertain quantity omitted from variance	Include its first-order variance contribution
Heuristic existence/count weights implied healthy disposition	Single-defect prior does not infer defect absence	Label heuristics and disable unsupported healthy release

All fixes have a bounded supported claim. Statistical quality/OOD/verification tests do not guarantee detection of all mismatch or measured confidence coverage.

Fresh current-source results

Eight held-out engineering seeds per condition/policy, six observations, twelve shared diagnostic opportunities, and three policies produced 48 completed runs. Every run and failure is retained. The protocol was recorded before execution; source fingerprints were unchanged and rechecked afterward.

Condition	Legacy mean error	Nuisance-aware mean error	Risk-controller mean error
Nominal	16.04 mm	14.30 mm	13.35 mm
Mismatch plus noise	189.95 mm	144.36 mm	143.71 mm

Paired nuisance-aware advantages over legacy: 1.74 mm, 95% interval [-6.78,9.50] nominally; 45.59 mm, [-22.85,107.10] under mismatch. Paired risk advantages: 2.69 mm, [-0.78,7.59] and 46.24 mm, [-10.48,112.27]. All intervals cross zero. All policies succeed within 20 mm in 6/8 nominal cases. Under mismatch, legacy succeeds in 0/8 and each optional policy in 1/8. Large stressed errors and missed mismatch remain material failures. The optional policies stay off by default.

Timing is descriptive on a shared workstation: about 0.33-0.37 seconds per six-observation legacy case, 0.45-0.50 for nuisance-aware, and 20.84-20.87 for risk. This is not a dedicated real-time benchmark. Fixed budget cannot establish measurement compression. Software-control improvements are demonstrated by specific invariants, not by an unsupported claim of field accuracy.

Publication/IP deliverables

• Updated IEEE working manuscript: paper/main.tex and the generated October table.

• Technical disclosure and preliminary adjacent-art worksheet: docs/PATENT_REVIEW_PACKAGE.md and CURRENT_INVENTION_DISCLOSURE.md.

• Frozen protocol and raw per-run report: VALIDATION_PROTOCOL_2026_10_07.md and research_results/release_validation_2026_10_07.json.

• Local source/evidence ZIP with separate manifest, excluding private environment, credentials, raw data and binary models.

• Detailed readable audit/disclosure PDF and a rebuilt manuscript PDF.

No novelty, filing status, editorial acceptance or course outcome is inferred from these artifacts. Existing manuscript authorship was preserved; human inventorship and university ownership need real records. The closest literature/patents still require a professional claim/family search. A physical-control method must have an enabling apparatus description and evidence at the scope actually claimed.

Remaining release/claim limitations

1. No blind physical adaptive hidden-defect study; stress accuracy is not acceptable as a deployment claim. Real coupling, triggers/clocks, dispersion, boundaries and transfer functions remain uncharacterized for the intended apparatus.

2. One dominant location grid; existence, number, type, depth, radius and severity are not validated joint inverse outputs.

3. Different planning and waveform likelihoods; configured costs/trust/confidence are proxies/preferences rather than measured maintenance loss or field probabilities.

4. Shared operator key, cleartext trusted-LAN mode and per-process limits; multi-user/public deployment requires additional authentication/TLS architecture.

5. An emergency latch prevents future acquisition but cannot interrupt an in-flight hardware action. A physical interlock must be built and measured.

6. Hashes and imported manifests prove consistency only; no independent sensor authenticity, signed timestamp or human contribution is established.

7. Online npm audit remains pending explicit approval. Automatic approval review rejected sending dependency metadata to registry.npmjs.org. Public advisory checks and targeted patches do not constitute a complete transitive audit.

The strongest next milestone is the specimen/defect-held-out physical protocol in VALIDATION_PROTOCOL_2026_10_07.md. Reviewers should receive the genuine failure cases and this claim boundary alongside the functioning demo and software evidence.

ARGUS technical invention disclosure for professional review

Prepared 7 October 2026. Technical input, not a filed application, novelty finding, legal claim set, or patent-pending representation. Human conception, ownership, prior disclosures and jurisdiction are unresolved administrative facts.

Working title and technical field

Sequential acoustic interrogation with correlated propagation calibration, measurement-validity gating, and verification-supported termination.

The implemented field is operator-guided acoustic/vibration interrogation of a rectangular panel with one dominant localized scatterer. The objective is to choose physical measurements under uncertain propagation and finite action cost.

Problem addressed

Speed and instrument timing uncertainty can confound defect localization. Low-quality, aliased, incomplete or unsupported recordings can create misleading spatial concentration. A diagnosis-only controller can spend scarce measurements on structural ambiguity that cannot be resolved until propagation is calibrated. The program compares supported diagnostic and reference actions, gates the actual acquisition before updating structure, and requires a fresh qualifying verification observation before a confident termination.

Enabled software embodiment

1. Define a 0.60 x 0.40 m panel and 20 x 20 location grid. The synthetic effective speed prior is 180 m/s and timing prior is 0.8 ms. These are illustrative, not measured material properties.

2. Represent propagation speed and timing with means and a correlated 2 x 2 covariance. Other nuisance quantities remain separate approximate marginals.

3. Enumerate bounded source/receiver/waveform candidates within drive, duration, frequency, spacing and no-go restrictions. Live forward levels are analytical and physics-signature only. The default policy uses explicit ranking proxies.

4. Optionally compare diagnostic and direct-reference actions using expected squared position loss plus acquisition cost. Partition all spatial mass into regions; retain within-region spread. Gaussian quadrature branches

over at most two observations and conditions later arrivals using shared propagation covariance. A finite beam restricts optimization.

5. Execute one selected action via the simulator or an acquisition adapter. Validate original bandwidth, mono shape, finite samples, declared window, normalization and channel reliability before evidence fusion. Server-owned acquisition provenance selects the physical acoustic-reference guard.

6. For a direct reference, update propagation covariance from one scalar delay; do not count that delay as two independent observations or as defect evidence. Use varied reference distances to improve speed/timing identifiability.

7. For diagnostic evidence, subtract the nominal direct response, correlate with the excitation, sample support at each candidate bistatic lag, construct the generalized likelihood, then assess OOD against pre-observation belief. Block unsupported data and temper caution. Fuse accepted evidence in log space.

8. A confident stop requires the latest accepted, nominal, consistent verification measurement with changed geometry or excitation content. Amplitude-only, duration-only and reciprocal-probe changes do not count as fresh verification.

9. Preserve state and record the action, waveform, configuration, planner, diagnostics and transition hashes with the actual source fingerprint. An independently retained checkpoint may anchor history-consistency verification.

The firmware reference enables an ESP32 exciter driver/conditioned ADC contract, and adapters support WAV, local/browser microphones and phone/edge nodes. The repository does not establish an assembled, synchronized, characterized NDE apparatus; its hardware embodiment requires measured construction evidence.

Mathematical support

For candidate location z, source S, receiver R, speed v and timing t0:

```
arrival(z,e) = t0 + (distance(S,z)+distance(z,R))/v
H = [-path/v^2, 1]
arrival_variance = H C H^T + R
K = C H^T / (H C H^T + R/quality_weight)
new_mean = mean + K (observed_reference_delay - predicted_delay)
new_C = (I-KH) C (I-KH)^T + K(R/quality_weight)K^T
```

With region centroid c_h, mass p_h and within-region second moment w_h:

```
position_risk = sum_h p_h (||c_h||^2 + w_h) - ||sum_h p_h c_h||^2
candidate_value = prior_risk - expected_terminal_risk - acquisition_cost
structural_posterior_next = normalize(prior * likelihood^evidence_weight)
```

The default cost conversion is 0.0001 m^2 per declared research cost unit. It is a configured preference, not a measured maintenance loss. The waveform inference likelihood differs from the regional Gaussian planning model. This is an acknowledged model-consistency and physical-validation gap.

Claim-support worksheet

Candidate element	Implemented support	Excluded or unresolved
Sequential feasible acoustic action selection	Engine, candidate planner, hardware API and recorded histories	Global optimality, robotics, arbitrary modalities
Correlated reference calibration	Nuisance covariance, Joseph update, distinct reference path options and tests	Full hierarchical posterior, exact physics
Common physical-loss diagnosis/calibration comparison	Optional finite-horizon risk planner and synthetic evaluation	Guaranteed fewer measurements, superior field error
Pre-fusion validity and provenance controls	Source bandwidth/window/unit checks, channel/quality/OOD gates	Calibrated physical confidence or spoof-proof sensors
Fresh verification-supported stop	Latest-observation consistency and meaningful diversity tests	Field coverage/safety certification
Bound recorded state/evidence identity	Content-bound ledger and independent checkpoint option	Signatures, trusted timestamps, human authorship proof

Any draft claim should be tied to a reproducible technical sequence supported by these implementation files, not the general idea of AI or Bayesian mathematics. This worksheet is material for a practitioner to assess, not an asserted claim of novelty or non-obviousness.

Preliminary adjacent-art comparison

Reference	Relevant existing subject matter	Specific comparison still needed
EP2214009A2	Reference wave responses and probabilistic defect-location mapping	Whether the ordered diagnosis/reference competition and qualifying verification are disclosed or obvious
US9964468B1	Model/scenario-based structural-health sensor placement	Fixed arrangement versus live belief-conditioned sequential action selection and calibration
US20230401694A1	Confidence-driven inspection-location selection and model retraining	Wave-response discrimination and propagation-reference comparison rather than inspection-label acquisition
Flynn and Todd guided-wave placement literature (paper bibliography)	Bayesian sensor placement and value of information	Closest sequential calibration/dual-control methods, not merely fixed networks

The three patent documents were consulted again during this audit. This is a limited comparison, not an exhaustive family/claim search or freedom-to-operate analysis. Existing Gaussian conditioning, experimental design, wave localization, dual control and hash chains are not presented as newly invented principles.

Evidence that exists and evidence still required

Existing: executable code, security/scientific regression tests, seeded synthetic studies, separate public measured-data bridge artifacts, and source/evidence packaging. The fresh release study is documented separately and keeps all failures.

Required for efficacy claims: blind, specimen-held-out measurements on the actual apparatus with independent geometry/timing truth, appropriate comparison methods, coverage/false-confidence checks and matched-cost policy comparisons. The existing public GFRP classifier and one-plate impact experiments cannot establish adaptive persistent-defect localization on this apparatus.

Administrative and disclosure record to complete

• Actual human inventor(s), each claim-specific contribution and dated conception evidence; code authored with an AI tool is not itself a human conception record.

• University/credit-project ownership, collaborator agreements and any sponsor or competition obligations.

• Repository visibility history, presentations, uploads, demonstrations, offers, prior filings and exact disclosure dates.

• Target jurisdiction, suitable filing form, enabling drawings/measurements, and a professional independent-claim/family/literature comparison.

Public disclosure can affect filing options; the applicable rules must be checked for the chosen jurisdiction before external publication. The USPTO provisional guidance explains disclosure requirements and timing for that US filing route. The EPO simulation guidance distinguishes computer implementation from technical effects. Neither establishes ARGUS eligibility or novelty in India or elsewhere. No filing or external release is performed by this preparation package.

ARGUS security and acquisition contract

Updated 7 October 2026. This is a local operator research instrument, not a multi-tenant service or a hardware safety controller.

Start locally

```
python run_argus.py
```

The launcher binds the frontend and backend to 127.0.0.1. It prefers the project `.venv` interpreter when present. The usual frontend address is http://localhost:5173; the configured API port in this workspace is 8001. Other devices cannot access local mode. The backend also checks the peer address, Host header and exact browser origin, so alternate Uvicorn bind flags do not silently convert local mode into anonymous LAN control.

Explicit shared LAN mode

```
python run_argus.py --lan
```

The launcher generates a per-run access key unless ARGUS_API_TOKEN is provided, sets exact hosts/origins for its discovered LAN address, and prints the key only to the operator terminal. Enter it under **Shared engine access** on the main or phone page. The key is scoped to the API and browser tab, not placed in a URL, localStorage, a NEXT_PUBLIC variable, or the frontend child-process environment.

Direct backend configuration uses ARGUS_NETWORK_MODE=lan, ARGUS_BIND_HOST, ARGUS_API_TOKEN, ARGUS_ALLOWED_HOSTS and ARGUS_ALLOWED_ORIGINS. LAN configuration fails closed unless all required controls exist. Tokens contain 32-256 URL-safe characters. Browser requests use Authorization: Bearer; browser WebSockets carry the key in a non-echoed authentication subprotocol, never a query string. `/health` is public but returns only service/security mode information. API documentation requires an authorized client when authentication is enabled.

HTTP LAN mode is for a trusted network: it does not encrypt the key or data. Use a correctly configured TLS reverse proxy and exact HTTPS origins for hostile or shared networks. A shared key has operator-wide access; per-user roles, tenant isolation, institutional authentication and distributed rate limiting are not implemented. Default limits assume one ASGI worker.

The edge client reads ARGUS_API_TOKEN from its environment. It refuses HTTP redirects rather than forwarding credentials to another destination, registers once per connection, and acquires each newly observed recommendation. Keep the operator responsible for actual probe placement.

Modern phones require a trustworthy HTTPS context for microphone/camera access. The printed plain HTTP LAN page supports manual placement and engine connection; it explicitly disables unavailable capture. Localhost microphone/WAV acquisition and the laptop edge client remain alternatives. No sensor permission is bypassed.

Bounded inputs and execution

• HTTP JSON: 16 MiB; WAV file: 10 MiB; bundle file: 50 MiB, with bounded multipart overhead, body/chunk deadlines and work-concurrency/rate limits.

• WebSocket frame: 64 KiB before JSON parsing, with connection, message and idle limits. Authentication, Host and origin checks apply to sockets too.

• Original waveform: finite mono samples, adequate declared window, at most 384,000 samples and original-rate Nyquist support for the excitation band. Short captures are rejected. Longer captures may be explicitly cropped; resampling and window provenance are recorded. Missing evidence is not padded.

• Client sensor labels cannot disable physical reference/OOD checks. Acquisition provenance is chosen by the server adapter.

• Non-normalized units require explicit finite affine normalization and a named reference. This records a declaration, not a verified physical transfer function.

• Candidate, beam, rollout, cache and acquisition allocations are bounded. Metadata is JSON-only, finite, size/depth bounded and rejects unknown fields.

• Device paths are restricted to discovered serial ports. Per-device locks serialize access. Emergency latch, no-go geometry and drive limits are checked before physical acquisition begins.

The UI dispatches emergency-stop requests even during a pending acquisition and preserves an acknowledged latch against late responses. An already-running hardware action is not interrupted: session/device locks span acquisition. A real apparatus requires an independently tested physical emergency interlock.

Models, bundles and evidence

The current trainer writes pickle-free ZIP/NPY/JSON forward checkpoints. Readers preflight member counts, expanded size, paths, NPY dimensions/dtypes, hashes, finite weights and normalization scales. Historical torch/pickle checkpoints are refused without deserializing or silently converting them. Retrain using the current trainer. Safe round-trip prediction is covered by regression tests.

Imported bundles and replay banks are checked before state hydration. Bounded zlib decoding prevents stored-waveform decompression from growing without a limit. Replay requires sealed truth and recorded compatible actions; explicitly labelled reference recordings alone may be used as calibration evidence.

Ledger schema 2 binds planner and diagnostic metadata as well as waveform, analysis and posterior content. Software identity includes dirty and untracked source bytes, not just Git HEAD. Hashes establish consistency, not authorship, sensor authenticity or trusted time. Retain an independent checkpoint if full history replacement is in scope.

Dependency verification boundary

Next.js was patched to 16.3.8 following the publisher's September 2026 release. The project virtual environment uses PyTorch 2.6.0+cpu, which closes the published <=2.5.1 checkpoint-loading issue. Pickle-based readers were removed independently of that package update. The newest tested CPU build failed Windows DLL integration; the working release was deliberately pinned to the compatible patched version.

Online npm audit was rejected by automatic approval review because sending dependency metadata requires explicit user permission. Until that permission is received, no complete transitive-advisory audit is claimed. Public advisories and local lockfile/source checks informed the fixes; an unperformed audit is not reported as zero vulnerabilities.

Sources: Next.js publisher release, PyTorch publisher advisory GHSA-53q9-r3pm-6pq6.

ARGUS publication and academic-review readiness

Prepared 7 October 2026. The local package supports a transparent prototype/method review; it does not establish a publication acceptance, patent grant or course grade.

Contribution stated at the supported scope

ARGUS is an inspectable single-scatterer acoustic interrogation prototype combining conditional spatial inference, uncertain propagation calibration, finite candidate selection, acquisition-validity/OOD gating and verification-supported termination. An optional two-observation regional arrival-risk controller compares diagnosis and direct-reference calibration in common squared-position units. The hardened system records actual source identity and content-bound evidence.

Do not describe the contribution as a new Bayesian update, globally optimal design, fully joint learned inference, physical depth/type/size reconstruction, a certified instrument, or a proven accuracy upgrade. The false-certainty and input-integrity controls are concrete engineering results, but efficacy remains an empirical question.

Manuscript corrections

The IEEE-format working manuscript in paper/main.tex now distinguishes:

• Separate structural and metrology state from an exact joint posterior or deployed particle inference.

• The executed quality/channel/OOD evidence exponent from a conceptual trust/temperature formula that is not used by the runtime.

• Fresh qualifying verification from threshold-only stopping.

• KL-derived separation proxies from actual Jensen-Shannon divergence.

• Finite waveform candidates and live forward levels 0/1 from proposed all-fidelity continuous optimization.

• Fast geometric route reranking from actual optional observation branching <=2.

• Unchanged attribute priors and heuristic damage-existence weights from learned defect existence/type/depth/severity.

• September synthetic results from the source-fingerprinted October release run.

Numerical tables preserve adverse and inconclusive results; none is relabelled as physical evidence. Final test counts and limitations are in RELEASE_READINESS_REPORT.

Review 1 evidence map

Review topic	Concrete artifact	Honest next milestone
Problem and objective	README, current architecture and study guide	Define target specimen/apparatus and tolerances
Physics and method	Simulation, likelihood, covariance calibration, risk planner	Independently measure propagation/model error
Technical implementation	API/UI/device contracts and reproducible launcher	Assemble and characterize actual acquisition
Software correctness/security	Regression suite and JUnit report	External audit under a declared deployment threat model
Experimental evidence	Historical studies plus frozen October protocol/results	Blind physical study with matched budgets
Differentiation/IP	Technical disclosure and preliminary adjacent-art worksheet	Practitioner claim chart and human contribution record

Submission decisions still outstanding

Identify the venue and paper category; check its template, scope, authorship, AI-assistance disclosure and reproducibility/data requirements. Complete genuine author affiliations and contributions without inventing coauthors or experimental work. Obtain institutional approvals where applicable. A prototype/method paper can report limitations and failed stress cases, but acceptance is an editorial decision. No public upload, journal submission or patent filing was made.

Reproduce and inspect the local package

```
.venv\Scripts\python.exe -m pytest backend\tests -q -o addopts=""
  --junitxml=output/release/backend-tests.xml
cd frontend
npm.cmd test
npm.cmd run typecheck
npm.cmd run lint
npm.cmd run build
```

From the project root, run the frozen validation command in VALIDATION_PROTOCOL_2026_10_07.md. Build and verify the local source/evidence ZIP using scripts/build_release_bundle.py. The manifest excludes private environment files, credentials, raw external data and binary models. Hashes show which files were reviewed; they are not signatures or proof of scientific correctness.

ARGUS 7 October 2026 release validation protocol

This is a fixed, synthetic engineering validation of the hardened working tree. It is not a physical trial, a powered efficacy study, or a patent-novelty test. The protocol was recorded before executing this release's numerical run.

• Seeds: 64000-64007, not the September development or validation seeds.

• Conditions: nominal; separately generated acquisition with velocity x1.02, attenuation x1.04, and noise x2.0 relative to the inference model.

• Policies: legacy_proxy, nuisance_aware, risk_calibration. Each uses the same pre-generated feasible diagnostic opportunities for a paired scenario.

• Budget: six observations per policy, twelve diagnostic opportunities per step. Fixed budget deliberately does not establish measurement compression.

• Outcomes: attempted/completed runs, execution failures, physical MAP error, posterior entropy, empirical R90 coverage, rejection and abstention, and the complete per-run trajectories retained by the evaluator.

• Uncertainty: paired case bootstrap, 10,000 resamples with the evaluator's fixed seed, percentile 95% intervals. Report intervals crossing zero as inconclusive.

• Timing: shared Windows workstation, not dedicated real-time hardware. Report computation as descriptive engineering data, not a deployment guarantee.

• Truth: used by acquisition and final scoring, not by candidate selection.

• Source boundary: hash scripts/research_validation.py and every backend/app Python file before execution; refuse publication use if source changes during the run. Preserve configuration and command in the artifact.

• No tuning after looking at these outcomes, no seed removal, no substitution of simulated missing physical responses, and no relabeling of evidence source.

Command:

```
.venv\Scripts\python.exe scripts\research_validation.py --seed 64000 --cases 8
  --steps 6 --candidates 12 --variants legacy_proxy nuisance_aware risk_calibration
  --output research_results/release_validation_2026_10_07.json
```

The September artifacts remain historical. Their numerical results must not be attributed to this hardened source tree. A successful software regression or a low-error single-seed demo cannot establish field defect-localization accuracy.

Physical closure study

Use the actual exciter, receiver, trigger, fixture and panel. Retain independently measured specimen/defect geometry, healthy reference recordings, source sampling rate, excitation window, normalization/transfer calibration and environmental metadata. Acquire all candidate outcomes required for offline policy replay, not only outcomes chosen by ARGUS. Hold out entire specimens or defect locations, keeping all related repeats/channels together. Compare adaptive and fixed/random policies under equal action opportunities and costs, retaining failures.

Primary physical outcomes must be declared before acquisition: position error vs budget, success at a justified tolerance, empirical containment coverage, false-confidence and rejection/abstention, plus comparison with an appropriate independent inspection/reference method. A representative study of damage existence and minimum detectable size requires additional healthy/defective labels; the present one-scatterer locator cannot substitute for it.